

CUSTOMER ENGAGEMENT & PRODUCT UTILIZATION ANALYTICS FOR RETENTION STRATEGY USING POWER BI

Project Report

Submitted By

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Technology Used

- Microsoft Power BI
 - Power Query
 - DAX (Data Analysis Expressions)
 - Microsoft Excel
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Project Domain

Data Analytics & Finance

Submitted To

Unified Mentor Private Limited

ABSTRACT

This project focuses on analyzing customer engagement behavior, product utilization patterns, and relationship strength indicators to understand customer retention and churn behavior in the European banking sector.

The dashboard includes KPI cards, churn analysis, engagement segmentation, product utilization analysis, high-value customer risk detection, scatter plot diagnostics, retention strength scoring, and relationship strength index (RSI) evaluation.

Interactive filters such as engagement filters, product count sliders, balance thresholds, salary thresholds, and churn segmentation enable dynamic customer retention analysis.

The project demonstrates how Business Intelligence tools can transform raw banking customer data into actionable insights that support engagement-driven retention strategies, customer loyalty improvement, product optimization, and churn reduction.

INTRODUCTION

Customer retention has become one of the most critical challenges in the banking industry. Traditional customer analysis often focuses on demographics and financial strength, but modern banking analytics emphasizes customer engagement and relationship depth as major retention drivers.

This project develops a Customer Engagement & Product Utilization Analytics Dashboard using Power BI to provide actionable insights into:

- Customer churn behavior
- Product utilization impact
- Customer engagement analysis
- High-value customer risk detection
- Retention strength measurement
- Relationship depth analysis
- Premium customer loyalty patterns

The project enables data-driven retention strategy design and customer relationship optimization.

PROBLEM STATEMENT

Banks often possess extensive customer data but lack visibility into behavioral drivers of customer churn and loyalty. Financially strong customers may still leave the bank due to low engagement, limited product adoption, or weak relationship depth.

The organization faces challenges in identifying:

- Which engagement behaviors reduce churn
- Whether product depth improves customer retention
- Which premium customers are at risk
- Whether active customers show stronger loyalty
- Which customer segments are most stable
- How relationship strength affects retention

As a result, customer retention strategies may become generic and ineffective.

OBJECTIVES

- To evaluate the relationship between engagement and churn
 - To analyze the impact of product utilization on retention
 - To identify high-value disengaged customers
 - To measure relationship strength using RSI scoring
 - To identify strong and weak customer segments
 - To analyze churn patterns across geography and engagement levels
 - To develop an interactive Power BI dashboard
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TECHNOLOGIES USED

Technology	Purpose
Microsoft Power BI	Dashboard creation and visualization
Power Query	Data cleaning and preprocessing
DAX	KPI calculations and analytical measures
Microsoft Excel	Dataset management

DATA CLEANING AND PREPROCESSING

The dataset was cleaned and validated before analysis.

Data preparation steps included:

- Removed duplicate customer records
- Handled missing and null values
- Corrected data types
- Standardized categorical fields
- Validated binary fields
- Created engagement segments
- Created RSI scoring model
- Created churn classification columns
- Applied DAX calculations for KPI measures

Additional calculated columns included:

- Activity Status

- Churn Status
 - Engagement Segment
 - RSI (Relationship Strength Index)
 - RSI Segment
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EXPLORATORY DATA ANALYSIS (EDA)

1. KPI Metrics

The dashboard includes the following key performance indicators:

- Overall Churn Rate
 - Engagement Retention Ratio
 - Product Depth Index
 - High-Balance Disengagement Rate
 - Credit Card Stickiness Score
 - Average RSI
 - Strong RSI Customers
 - Weak RSI Customers
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2. Engagement vs Churn Analysis

Customer activity behavior was analyzed to understand its impact on churn.

Findings showed that inactive customers exhibit significantly higher churn rates compared to active customers. Engagement classification also revealed that active engaged customers demonstrate stronger loyalty and lower churn probability.

3. Product Utilization Analysis

Customer retention was analyzed based on the number of banking products used.

Single-product customers showed the highest churn rates, while customers using multiple products demonstrated stronger retention and relationship stability.

Product depth analysis confirmed that cross-product adoption improves customer loyalty.

4. Financial Commitment vs Engagement Analysis

Balance and salary cross-analysis was performed using scatter plot visualization.

The analysis identified several high-balance customers who still churned despite strong financial commitment. This indicated that financial strength alone does not guarantee customer loyalty.

The project also identified at-risk premium customers characterized by:

- High balances
 - Low engagement
 - Weak relationship depth
 - Increased churn probability
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5. Retention Strength Assessment

A Relationship Strength Index (RSI) model was developed using:

- Activity status
- Product utilization
- Tenure
- Credit card ownership

Customers were segmented into:

- Strong RSI
- Moderate RSI
- Weak RSI

Weak RSI customers demonstrated significantly higher churn rates, while strong RSI customers showed improved retention stability.

BUSINESS INSIGHTS

- Inactive customers have substantially higher churn rates
 - Product utilization strongly influences customer loyalty
 - Single-product customers are highly vulnerable to churn
 - Multi-product customers exhibit stronger retention behavior
 - High-balance customers may still churn due to weak engagement
 - Germany shows higher churn rates compared to other regions
 - Weak RSI customers contribute disproportionately to churn
 - Relationship depth is a stronger retention indicator than balance alone
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BUSINESS RECOMMENDATIONS

- Improve customer engagement initiatives
 - Develop loyalty and retention programs
 - Promote multi-product adoption strategies
 - Monitor inactive high-balance customers
 - Implement personalized retention campaigns
 - Strengthen relationship management for premium customers
 - Develop predictive churn monitoring systems
 - Increase cross-selling opportunities
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ADVANTAGES OF THE PROJECT

- Interactive and user-friendly dashboard
 - Engagement-driven retention analysis
 - Improved business decision-making
 - Identification of at-risk premium customers
 - Dynamic customer segmentation
 - Better retention strategy planning
 - Advanced relationship strength analytics
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LIMITATIONS

- Dataset is static
 - No real-time banking integration
 - Limited predictive analytics implementation
 - No machine learning churn prediction
 - External behavioral factors not included
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FUTURE SCOPE

- AI-based churn prediction
- Machine learning retention forecasting
- Real-time banking data integration
- Customer lifetime value prediction
- Cloud-based deployment
- Advanced behavioral analytics
- Automated customer risk alerts

CONCLUSION

The Customer Engagement & Product Utilization Analytics for Retention Strategy project successfully provides a comprehensive Business Intelligence solution for banking customer retention analysis.

Using Power BI, Power Query, and DAX, the project transforms raw banking customer data into actionable insights related to customer engagement, product utilization, retention strength, and premium customer risk.

The dashboard enables better strategic decision-making and supports engagement-driven retention optimization through behavioral analytics and relationship strength evaluation.

The project demonstrates that customer engagement and product depth are stronger indicators of retention than financial strength alone.

REFERENCES

- Microsoft Power BI Documentation
 - DAX Documentation
 - Power Query Documentation
 - Banking Analytics Concepts
 - Customer Retention Analytics Concepts
 - European Bank Customer Dataset
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APPENDIX

Features Implemented

- KPI Cards
- Engagement vs Churn Analysis
- Product Utilization Analysis
- RSI Scoring Model
- Retention Strength Analysis
- High-Value Customer Detection
- Scatter Plot Diagnostics
- Interactive Filters
- Product Count Slider

- Balance Threshold Slider
- Salary Threshold Slider
- Geography Analysis
- Churn Segmentation
- Gauge Chart Visualization
- Relationship Strength Index (RSI)